Question	Answer	Marks
1	**Defining the problem**	
	vary θ and measure d **or** θ is the independent variable and d is the dependent variable	**1**
	keep z constant	**1**
	Methods of data collection	
	labelled diagram of workable experiment including: • sheet supported with stand or block(s) • path of ball (from sheet to bench) **or** distance d indicated • at least two labels from ball, bench, sheet, block/clamp/stand/support	**1**
	description of method to determine position of ball as it makes contact with the bench, e.g. paint ball, sand, video and playback in slow motion / frame by frame (with rule in view)	**1**
	measure d with a rule **and** measure z with a rule or calipers	**1**
	measure θ with a protractor **or** determine θ using (a rule for) appropriate distances with correct trigonometric relationship	**1**
	Method of analysis	
	plot a graph of d against $\sin(4\theta)$ or equivalent Do not accept logarithms.	**1**

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Question	Answer	Marks
1		

d against $\sin(4\theta)$	$\sin(4\theta)$ against d
$P = \dfrac{g \times \text{gradient}}{v^2}$	$P = \dfrac{g}{v^2 \times \text{gradient}}$

1

d against $\sin(4\theta)$	$\sin(4\theta)$ against d
$Q = \dfrac{y\text{-intercept}}{\sqrt{z}}$	$Q = -\dfrac{Pv^2 \times y\text{-intercept}}{g\sqrt{z}}$ **or** $Q = -\dfrac{y\text{-intercept}}{\text{gradient} \times \sqrt{z}}$

1

Additional detail including safety considerations **6**

D1 precaution linked to prevent <u>ball rolling</u> (off surface), e.g. use of cushion / sand box / barrier <u>to stop ball</u> (rolling onto the floor)
or
precaution linked to (bouncing) <u>ball</u> or <u>sand (</u>spray) e.g. use of goggles <u>to protect eyes</u>

D2 keep v <u>constant</u>

D3 method to keep v constant, e.g. keep the distance the ball falls constant to <u>keep v constant</u>

D4 method to determine v or v^2, e.g. use of
$v = \sqrt{2g \times (\text{height ball falls})}$
or
$v^2 = 2g \times (\text{height ball falls})$

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Question	Answer		Marks
1	D5	set square correctly positioned between rule and bench to ensure that rule is vertical to measure ***z*** or determine initial height of ball	
	D6	method to ensure ***z*** is constant, e.g. as θ is changed, (re-)mark point of contact or change release point of ball	
	D7	description of method to ensure ball is released above point of contact on sheet, e.g. use a plumb line from the ball through point of contact with sheet or (vertical) rule (and set square)	
	D8	drop ball from large height to obtain greater values of *d*	
	D9	repeat experiment for each θ **and** average *d*	
	D10	relationship valid <u>if</u> a straight line produced (with *y*-intercept of e.g. $\left(Q\sqrt{Z}\right)$). Do not accept line passing through the origin.	

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